

STAT 6910 Advanced R

Homework 2

by

Jürgen Symanzik

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UTAH STATE UNIVERSITY

Logan, UT

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1 General Instructions

You have to work in small groups of exactly two students on this assignment. Unless you have a strong request to work with someone else for this assignment, you will work with the same partner. Any requests to work with a new group partner must be made by 10pm on Monday, November 11, 2013.

I would suggest that you first work independently on this assignment and then compare your results with your partner. You can submit only one final version!

You have to submit two files via e-mail: The `.Rnw` file that contains the $\text{\LaTeX}$ and R/Sweave (or R/knitr) code and the resulting `.pdf` file prepared on your side. Your files should be named `LastName1_LastName2_HW02.Rnw` and `LastName1_LastName2_HW02.pdf`.

Your R code has to be fully reproducible! Make sure to reset the random number seed once. You cannot include any constants in your $\text{\LaTeX}$ text. Rather, you have to work with `\Sexpr` to extract numeric values from your R code. I will likely make changes to your R code, e.g., change some of the numerical values, to check whether your results properly get adapted to these new values.

Use this `.Rnw` file as the basis for your answers. Update the title page, e.g., fill in your names and indicate when this homework gets submitted. Moreover, fill in the chunks of R code where the current file states `### Your R code here`. Before you submit your homework, check once more that you follow all recommendations from Google's R Style Guide (see <http://google-styleguide.googlecode.com/svn/trunk/Rguide.xml>).

For general questions related to this homework, please use the corresponding discussion board in Canvas! I will try to reply as quickly as possible. Moreover, if one of you knows an answer, please post it. It is fine to refer to web pages and R commands, but do not provide the exact R command! This will be the task for each individual group.

2 Question 1 (8×8 Points = 64 Points + 4 EC Points)

Assume we are interested in properties of the following statistical process: First, a n -sided die ($n \geq 2$) is rolled once (see Figure 1 for examples of such dice). Suppose the outcome is k , where $k \in \{1, \dots, n\}$. Now, this same n -sided die is rolled k times. We are interested in the distributional properties of a random variable called D that represents the **sum** of these k dice rolls (but ignoring the outcome of the very first die roll in the calculation of the sum).



Figure 1: Examples of 4-, 6-, 8-, 10-, 12-, and 20-sided dice, obtained from <http://jackadreams.info/2012/11/25/dice-divination/>.

1. Write an R function that simulates a n -sided die ($n \geq 2$) and then determines the sum of k dice rolls of the same n -sided die. The return value of this function should be just this sum.

> *### Your R code here*

2. Write a function that simulates m of these two-staged dice rolls. For the following parts, work with $m = 10,000$. However, to speed up the calculations and the translation of your R/Sweave/L^AT_EX report, reduce m initially.

> *### Your R code here*

3. Provide at least two different graphical summaries of your experiment for $n = 4$ and $n = 6$ that show the sampling distribution of the sum of the dice rolls. Keep in mind that these are discrete distributions — so do not over-simplify your plots. Fine-tune your figures for the resulting pdf version and not for the screen version.

> *### Your R code here*

4. What are $E(D)$ and $Var(D)$, based on your m simulation runs for $n = 4$ and $n = 6$? And what is a 90% confidence interval for the mean of the sums?

> *### Your R code here*

5. *Manually* tabulate all possible ordered sequences in a $\text{\LaTeX}$ table when $n = 2$ and $n = 3$. There should be 6 and 39 such sequences, respectively. For each sequence, add two additional columns that list the sum d and the probability p for each of these sequences. Indicate the probabilities as a fraction and not as a decimal. Then, create two more tables that only list the sums d and the probabilities for these sums. Check in each table that the probabilities sum up to exactly 1.

6. What is the **exact** result for $E(D)$ and $Var(D)$ for $n = 4$ and $n = 6$?

To obtain these results, generalize your hand calculations from the previous part and write an R function to obtain these tables and probabilities. Include these computer-generated results here, but only for $n = 2$ and $n = 3$. Compare these results with those from your manual calculations and make sure they are matching. If not, check whether you made a mistake in your manual calculations or in your R code.

Then, use simple summation (as introduced in any intro stats course) to determine $E(D)$ and $Var(D)$ for $n = 4$ and $n = 6$. Be careful that you work with the proper probabilities.

> **### Your R code here**

7. Determine $P(D < 5)$ and $P(12 \leq D \leq 16)$ for $n = 4$ and $n = 6$, based on your simulation results. Also use the results from the previous part to obtain these exact probabilities. Be careful with the end points! Create a table in $\text{\LaTeX}$ that lists the simulation and exact results for these probabilities, the absolute differences ($| \text{simulated} - \text{exact} |$), and the relative differences ($\text{simulated} / \text{exact} \cdot 100\%$). Round to two decimal digits.

> **### Your R code here**

8. Summarize your results. Was $m = 10,000$ a good choice, or would it be (highly) beneficial to increase m ?
9. (4 EC Points) Find a solution based on Mathematical Statistics for $E(D)$ and $Var(D)$ for $n = 4$ and $n = 6$ and write it up in $\text{\LaTeX}$. If you are interested in my $\text{\LaTeX}$ lecture notes for Stat 6710/6720 as a template, I can send those upon request.

3 Question 2 (5 × 8 Points = 40 Points)

Obtain a Monte Carlo estimate for the base of the natural logarithm, e . This question is based on the concept we used to find an estimate of π via simulation. We do **not** want to calculate e via any of the mathematical formulas we know from Mathematical Statistics here. Your results must be fully reproducible!

As a reminder, it holds that

$$\int_1^e \frac{1}{x} dx = \ln(e) - \ln(1) = 1 - 0 = 1.$$

Sketch a graph that shows this fact and use this as the basis of your simulation. You can assume that $e \leq 4$.

Hint: You need to find the smallest of the sorted x -values such that the area underneath the curve $1/x$ inside the rectangle $[1, x] \times [0, 1]$ is ≥ 1 .

1. Your estimate of e should be based on $m = 1,000$, $m = 10,000$ and $m = 100,000$ pairs of random numbers. First write a function that creates a single pair of these random numbers. Then write a function that calls your first function m times. The return value of this function should be a single estimate of e .

> **### Your R code here**

2. Create plots of the first four simulation runs for $m = 1,000$ that highlight the points that contribute to the calculation of e , the area of interest, and the estimate of e for that specific simulation run. Optimize the spacing and size of these plots and arrange them in a single figure.

> **### Your R code here**

3. Obtain results for 100 and 1,000 replications for the three different values of m , i.e., you are conducting $2 \times 3 = 6$ different simulation experiments. Tabulate the mean, median, and a 95% confidence interval for each of these six experiments. How close do you get to the true e ?

Also, create six boxplots in a 2×3 layout. Label them clearly. Recall the principle for small multiples that must be applied here.

Note: 1,000 replications of $m = 100,000$ pairs will take a while. Test your code (R & L^AT_EX) initially for smaller values and only use the larger numbers when everything seems to be correct.

> **### Your R code here**

4. Tabulate results from the expression

$$\left(1 + \frac{1}{n}\right)^n,$$

for $n = 100, 1,000, 10,000$, and $100,000$ and compare with your simulation results. Which seems to be the better way to obtain e — and why?

> ### Your R code here

5. This is a tricky question that requires some careful mathematical and geometrical considerations and not only some fast implementation of R code. In this last part, you have to provide a verbal explanation of your algorithm — and why it is expected to give us a correct result.

Note: As a reminder, make sure that your figures and tables are properly numbered and have a meaningful caption — and that you refer to each of them at least once from the main text.